# DESCENT OF FLAT FINITE-PRESENTATION MODULES ALONG LIMITS

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*Editorial note.* No omissions in the counted span: the reduction is worked and the algebra core is a precise citation of a proved lemma.

*Allowed standard prerequisites (editorial).*

- Limit-flat finite presentation. Algebra, Lemma flat-finite-presentation-limit-flat

*Source reference.* Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 04AI: sources/stacks-chow/tags.txt, source lines 5617–5617. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02JO: sources/stacks-chow/tags.txt, source lines 3343–3343. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *With notation and assumptions as in Lemma 01ZM. Let $i \in I$. Suppose that $\varphi_i : X_i \to Y_i$ is a morphism of schemes of finite presentation over $S_i$ and that $\mathcal{F}_i$ is a quasi-coherent $\mathcal{O}_{X_i}$-module of finite presentation. If the pullback of $\mathcal{F}_i$ to $X_i \times_{S_i} S$ is flat over $Y_i \times_{S_i} S$, then there exists an index $i' \geq i$ such that the pullback of $\mathcal{F}_i$ to $X_i \times_{S_i} S_{i'}$ is flat over $Y_i \times_{S_i} S_{i'}$.*

*Proof.* (This lemma is the analogue of Lemma 04AI for modules.) For $i' \geq i$ denote $X_{i'} = S_{i'} \times_{S_i} X_i$, $\mathcal{F}_{i'} = (X_{i'} \to X_i)^*\mathcal{F}_i$ and similarly for $Y_{i'}$. Denote $\varphi_{i'}$ the base change of $\varphi_i$ to $S_{i'}$. Also set $X = S \times_{S_i} X_i$, $Y = S \times_{S_i} X_i$, $\mathcal{F} = (X \to X_i)^*\mathcal{F}_i$ and $\varphi$ the base change of $\varphi_i$ to $S$. Let $Y_i = \bigcup_{j=1,\ldots,m} V_{j,i}$ be a finite affine open covering such that each $V_{j,i}$ maps into some affine open of $S_i$. For each $j = 1, \ldots m$ let $\varphi_i^{-1}(V_{j,i}) = \bigcup_{k=1,\ldots,m(j)} U_{k,j,i}$ be a finite affine open covering. For $i' \geq i$ we denote $V_{j,i'}$ the inverse image of $V_{j,i}$ in $Y_{i'}$ and $U_{k,j,i'}$ the inverse image of $U_{k,j,i}$ in $X_{i'}$. Similarly we have $U_{k,j} \subset X$ and $V_j \subset Y$. Then $U_{k,j} = \lim_{i' \geq i} U_{k,j,i'}$ and $V_j = \lim_{i' \geq i} V_j$ (see Lemma 01YX). Since $X_{i'} = \bigcup_{k,j} U_{k,j,i'}$ is a finite open covering it suffices to prove the lemma for each of the morphisms $U_{k,j,i} \to V_{j,i}$ and the sheaf $\mathcal{F}_i|_{U_{k,j,i}}$. Hence we see that the lemma reduces to the case that $X_i$ and $Y_i$ are affine and map into an affine open of $S_i$, i.e., we may also assume that $S$ is affine.

In the affine case we reduce to the following algebra result. Suppose that $R = \operatorname{colim}_{i \in I} R_i$. For some $i \in I$ suppose given a map $A_i \to B_i$ of finitely presented $R_i$-algebras. Let $N_i$ be a finitely presented $B_i$-module. Then, if $R \otimes_{R_i} N_i$ is flat

*Date*: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

over $R \otimes_{R_i} A_i$, then for some $i' \geq i$ the module $R_{i'} \otimes_{R_i} N_i$ is flat over $R_{i'} \otimes_{R_i} A$. This is exactly the result proved in Algebra, Lemma 02JO part (3). $\square$